Student 4 - Answer to Q1

Congestion control in TCP is mainly performed by the receiver, which tells the sender how fast it may transmit using the receiver window rwnd, and it is also related to flow control in the transport layer. In slow start the congestion window cwnd grows linearly by one MSS every round trip time, which is why it is called slow. When cwnd reaches the threshold ssthresh, congestion avoidance starts and the window then grows exponentially, doubling quickly so that bandwidth is used fully. If a timeout occurs the sender doubles ssthresh and keeps the same cwnd to maintain throughput. Fast retransmit means the sender waits for the timeout and then retransmits all packets, and fast recovery is the same as slow start after three duplicate ACKs. TCP also uses acknowledgement numbers, sequence numbers, checksums, three way handshake and sliding window protocols to provide reliable transport of data between hosts across routers over the internet, and all of these together form the congestion control approach in the network layer used by TCP and UDP.